H2 ANDERSON SERANGOON JUNIOR COLLEGE
HIGHER 2

2021 JC2 PRELIMINARY EXAMINATION

CANDIDATE NAME					
CLASS	2	1	/		

INDEX NUMBER		

H2 BIOLOGY 9744/03

Paper 3 Long Structured and Free-Response Questions **30 August 2021**

Candidates answer on Question Paper.
No Additional Materials are required. **2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.
Write in dark blue or black pen.
You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4 or 5	
	/75

This document consists of **23** printed pages and **1** blank page.

Section A

Answer **all** the questions in this section.

1 The development and function of an organism is largely controlled by genes. Any accidental changes to an organism's genes, in the form of gene mutations or chromosomal aberrations, can be potentially damaging.

(a) Compare gene mutation and chromosomal aberration.

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Gene mutations can affect the haemoglobin molecule, leading to sickle cell anaemia. HbA codes for the normal functional haemoglobin while HbS is the mutant recessive allele for sickle cell anaemia.

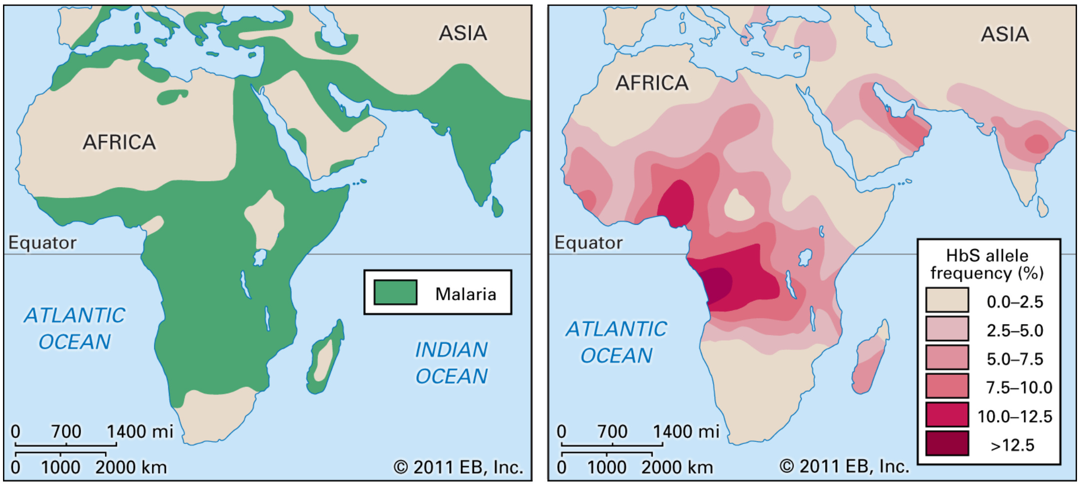

Fig. 1.1

(b) Comment on whether the data provided in Fig. 1.1 support a correlation between HbS allele frequency and incidences of malaria.

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Malaria is caused by the *Plasmodium* parasite. It has developed a wide range of defences against the human immune system, making an effective vaccine difficult to develop.

In 2011, randomised trials of a new vaccine called RTS were conducted. 894 children aged 5 to 17 months from Kenya and Tanzania were vaccinated with either the RTS vaccine or with a **placebo** (no active substance). Neither the children nor their families were told whether the vaccine or a placebo was administered.

The children were then monitored. Among those who were vaccinated, 40% fewer children developed malaria.

(c) **(i)** Describe the role of the placebo.

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(ii) Discuss the **economic** benefits of vaccination against malaria.

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(iii) One way to produce immunity to malaria in humans is to inject a vaccine consisting of irradiated malarial parasites that are unable to divide.

Assuming malaria parasites divide similarly to human cells, explain how the irradiation changes the pathogen so that it is unable to divide.

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In another study to test the ability of the RTS vaccine to protect against malaria, a clinical trial was conducted where recruited adult patients were randomly given either the vaccine or a placebo. They were then tested for antibodies against the malaria parasite after 10 weeks.

Table 1.1 shows the results of the trial.

Table 1.1

test group	number of adult patients	average amount of antibodies / arbitrary units
vaccine	10	3148
placebo	10	489

standard deviation $s = \sqrt{\dfrac{\sum(x-\bar{x})^2}{n-1}}$

t-test $t = \dfrac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}\right)}}$ $v = n_1 + n_2 - 2$

Symbols:
x = observed value
$\bar{x}$ = mean
n = number of observations (i.e. number of values within a data set)
v = degrees of freedom

<u>Table of *t* critical values</u>

degrees of freedom	probability, *p*				
	0.5	0.1	0.05	0.01	0.001
1	1.00	6.31	12.71	63.66	636.62
2	0.82	2.92	4.30	9.92	31.60
3	0.76	2.35	3.18	5.84	12.92
4	0.74	2.13	2.78	4.60	8.61
5	0.73	2.02	2.57	4.03	6.87
6	0.72	1.94	2.45	3.71	5.96
7	0.71	1.89	2.36	3.50	5.41
8	0.71	1.86	2.31	3.36	5.04
9	0.70	1.83	2.26	3.25	4.78
10	0.70	1.81	2.23	3.17	4.59
11	0.70	1.80	2.20	3.11	4.44
12	0.70	1.78	2.18	3.05	4.32
13	0.69	1.77	2.16	3.01	4.22
14	0.69	1.76	2.14	2.98	4.14
15	0.69	1.75	2.13	2.95	4.07
16	0.69	1.75	2.12	2.92	4.01
17	0.69	1.74	2.11	2.90	3.97
18	0.69	1.73	2.10	2.88	3.92
19	0.69	1.73	2.09	2.86	3.88
20	0.69	1.72	2.09	2.85	3.85

(iv) The calculated *t* value is 2.365.

Based on the information provided, explain if the vaccine is effective in conferring protection against malaria.

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For individuals who are carriers of the sickle cell anaemia allele, the best way to find out how sickle cell disease runs in their family is for a genetic counsellor to do a genetic test. Genetic information of the beta-globin alleles can be obtained through DNA sequencing, which is a modified Polymerase Chain Reaction (PCR).

DNA is extracted from the individual to be tested and used as a template in four independent PCR reactions. Each reaction contains the standard PCR components with dNTPs (deoxyribonucleoside triphosphates), and a different ddN (dideoxyribonucleoside triphosphate): ddA, ddG, ddC or ddT. Random incorporation of the ddN into the growing DNA chain results in chain termination. This generates a range of DNA strands of different lengths.

The DNA strands from each reaction are then **separated** and visualised using a DNA-staining dye. Each band corresponds to a DNA strand that has incorporated the ddN.

Fig. 1.2 shows how DNA sequencing can be carried out.

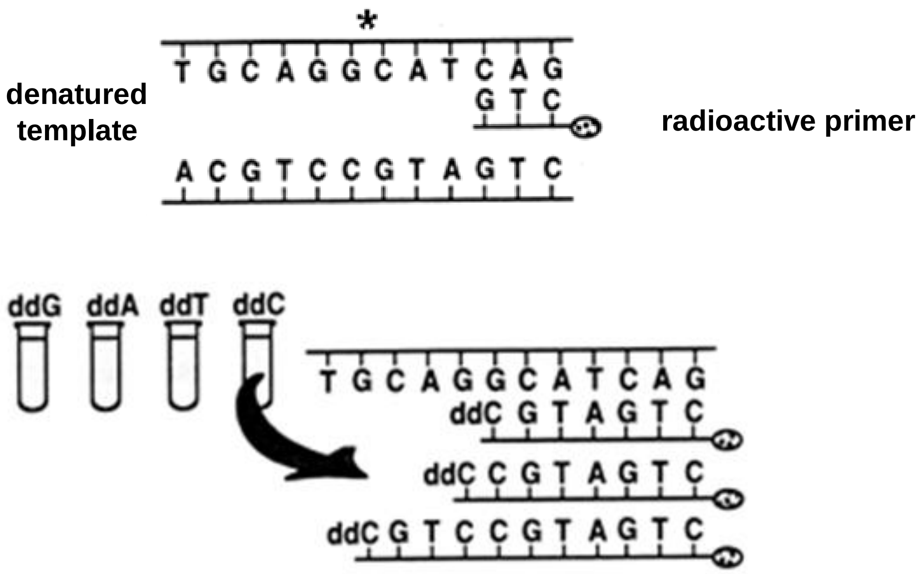

Fig. 1.2

Fig. 1.3 shows the structure of a ddN.

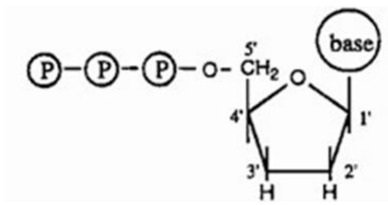

Fig. 1.3

The second procedure of the sequencing process produces a result shown in Fig. 1.4, from which the DNA sequence can be read.

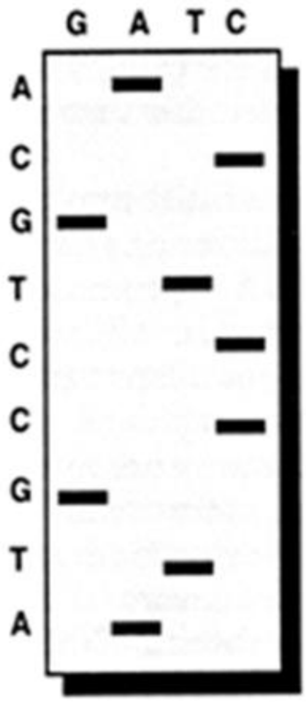

Fig. 1.4

(d) **(i)** On Fig. 1.2, label the 5' and 3' ends of the DNA strand marked with an asterisk (*). [1]

(ii) With reference to Fig. 1.3, explain how ddN results in the termination of chain elongation.

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(iii) With reference to Fig. 1.4, state the sequence of the longest strand synthesised. Indicate the 5' and 3' ends of the strand in your answer.

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(iv) Describe the procedure used to separate the DNA strands in Fig. 1.4.

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[Total: 23]

2 Fig. 2.1 shows the exchange of substances between a mitochondrion and a chloroplast in a leaf cell.

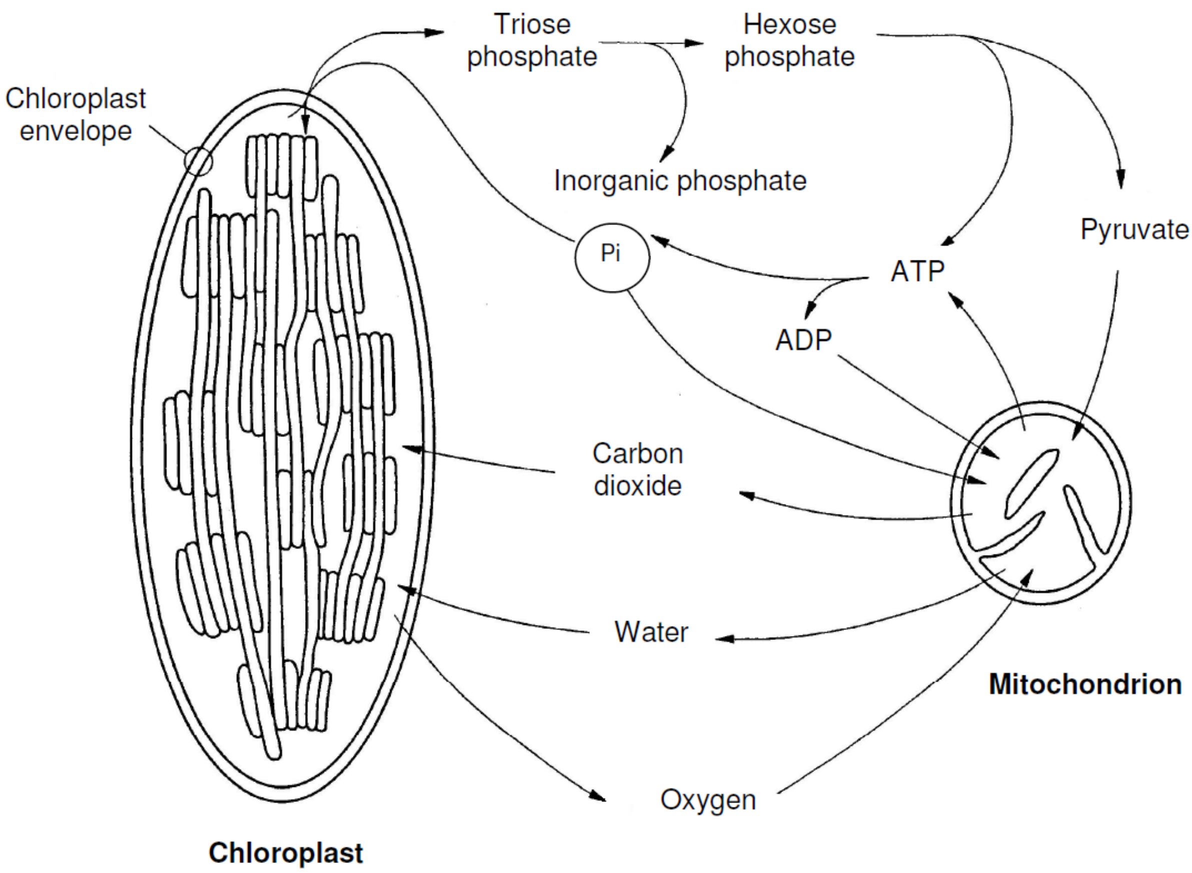

Fig. 2.1

(a) Explain how triose phosphate and hexose phosphate link the metabolic pathways between the chloroplast and the mitochondrion of this cell.

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(b) ATP is produced in both the chloroplast and the mitochondrion.

(i) Complete the table below by listing **all** the types of phosphorylation reactions occurring within these organelles that produce ATP.

organelle	type(s) of phosphorylation reaction occurring within
chloroplast	
mitochondrion	

[1]

(ii) Outline the fates of the ATP molecules produced in these organelles.

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(iii) An aqueous solution contains small membranous vesicles which are derived from a mitochondrial membrane. In these vesicles, the orientation of the ATP synthase enzymes is preserved as it is in the cell.

Fig. 2.2 shows four experimental systems **A**, **B**, **C** and **D** that are prepared in an attempt to synthesise ATP.

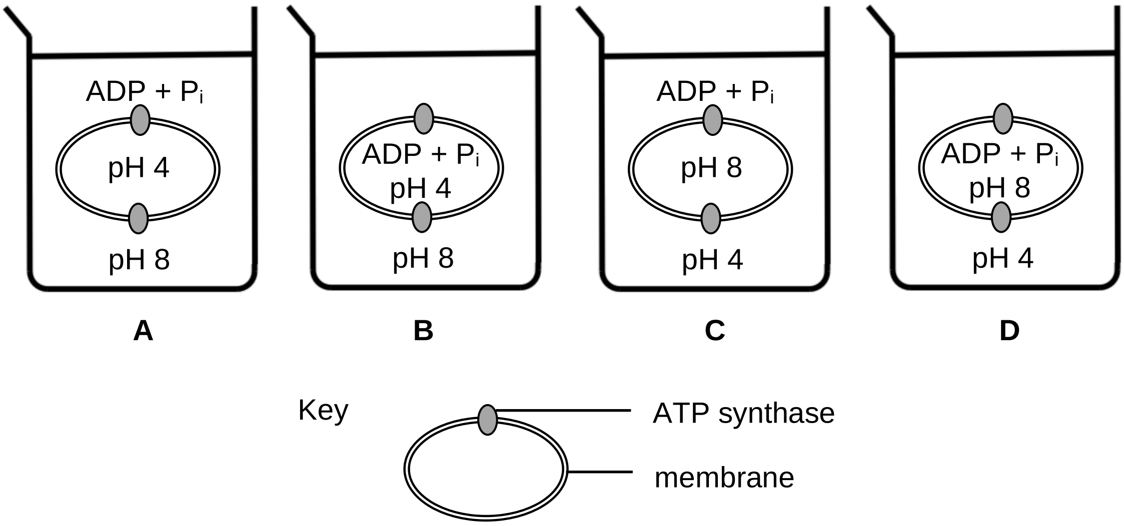

Fig. 2.2

Identify the experimental system that will result in the successful synthesis of ATP, and explain your choice.

Experimental system ……………………………………………………………….[1]

Explanation …………………………………………………………………………….

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Hexokinase is the enzyme involved in the first step of respiration. It catalyses the formation of G6P (glucose-6-phosphate).

ASR scientists recently invented inhibitor **X**, which can affect hexokinase activity.

An investigation was carried out to determine the effect on the production of G6P when the concentration of inhibitor **X** is increased. Five different concentrations (0 µM to 30 µM) of the inhibitor were used.

The concentration of G6P produced in each reaction mixture was determined at 10 minute intervals, up to 60 minutes.

The results are shown in Fig. 2.3.

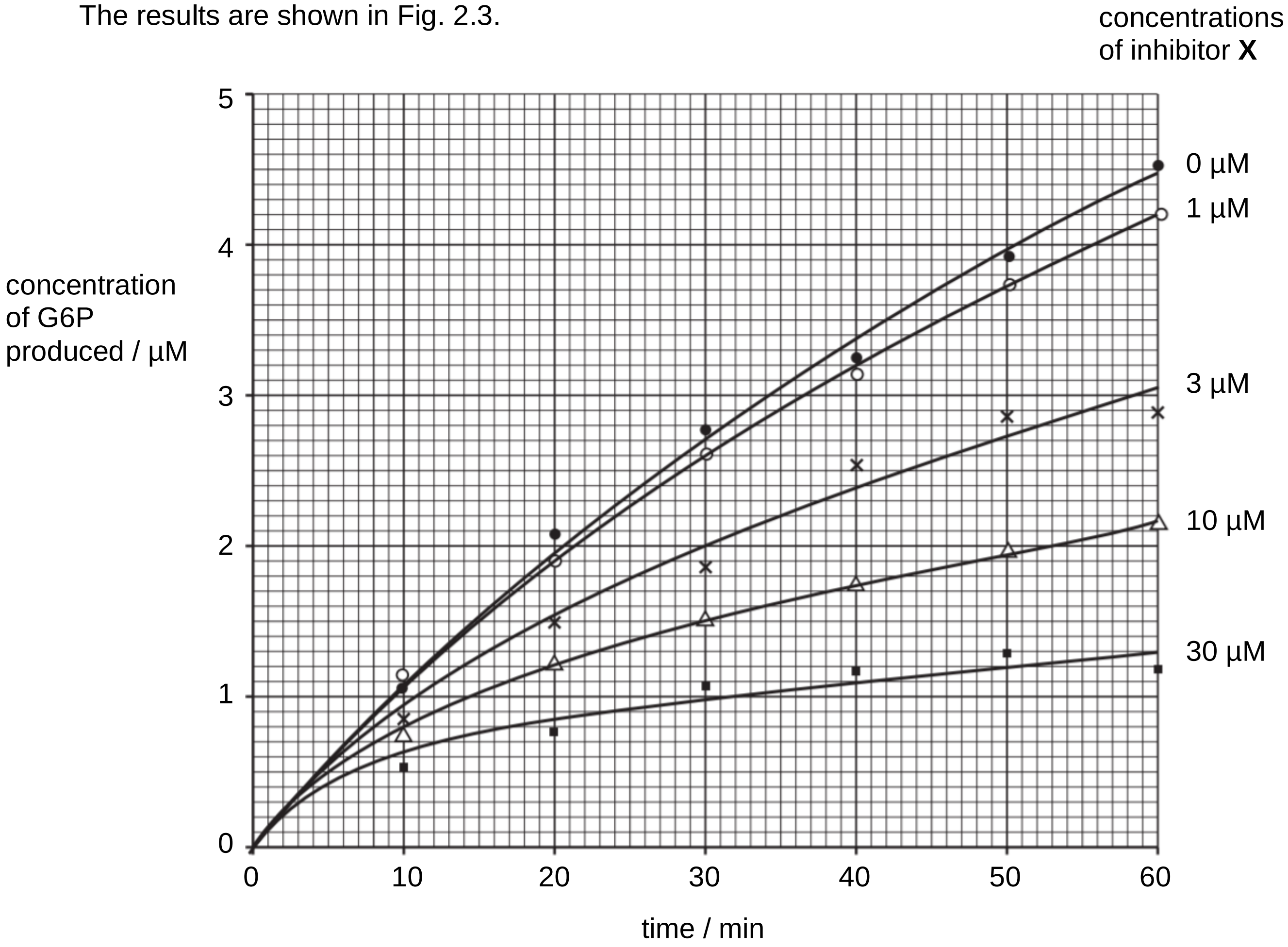

Fig. 2.3

(c) Use Fig. 2.3 to describe the effect of inhibitor **X** on the **rate** of G6P produced during the duration of the investigation.

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(d) From this investigation, it is **not** possible to conclude whether inhibitor **X** functions in a competitive or non-competitive manner.

Comment on the additional information needed to make a conclusion on how inhibitor **X** functions.

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3 *Salmonella enterica* is a species of pathogenic bacteria that causes gastrointestinal disease in humans and animals. The pathogen is shown to be remarkably adaptive, being able to invade a large range of host organisms.

Salmonella enterica serotype *typhimurium* (*S. typhimurium*) is the most significant cause of food poisoning in western countries. Infection is possible from ingestion of contaminated food and water. *S. typhimurium* typically enters the body via the gut. The intestinal barrier thus plays an important role in preventing infections by *S. typhimurium*.

Fig. 3.1 shows the composition of the intestinal barrier.

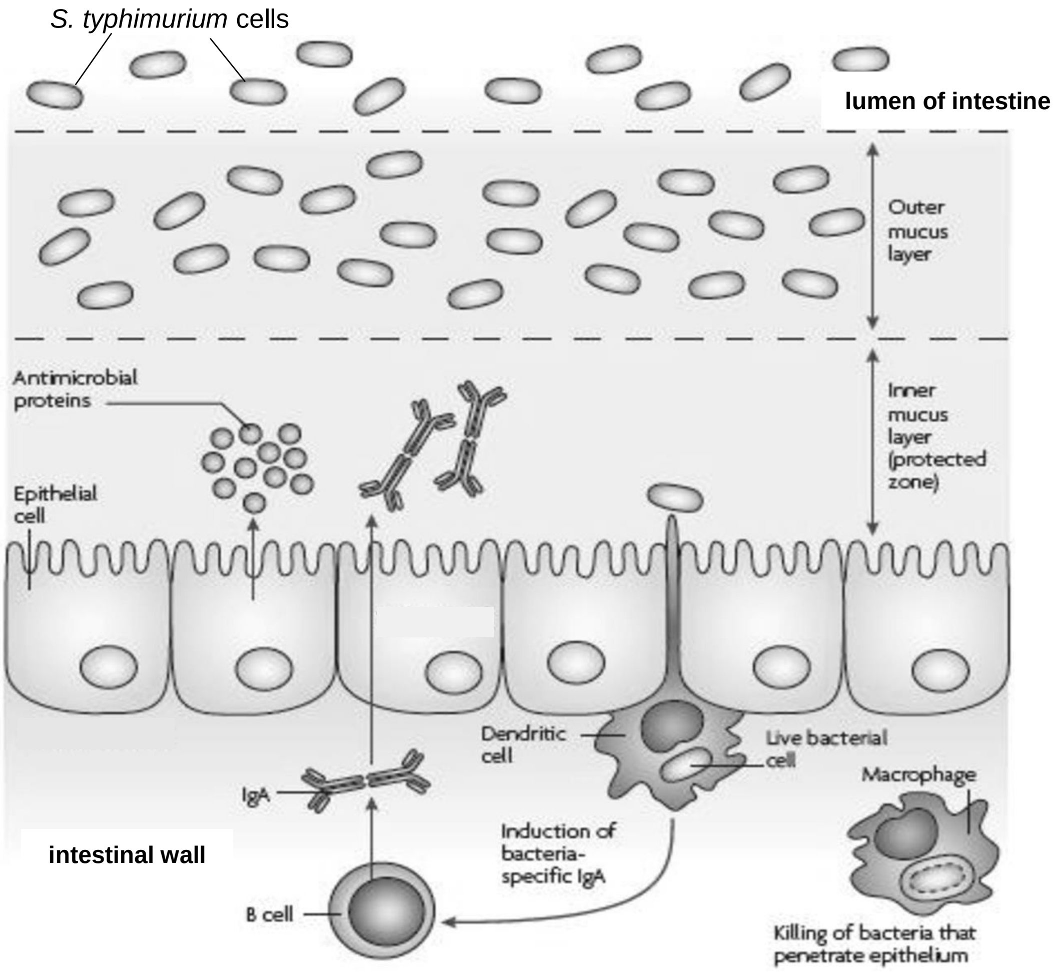

Fig. 3.1

(a) With reference to Fig. 3.1 and biological knowledge of the human immune system,

(i) describe the role played by the gut epithelial cells in preventing infections by *S. typhimurium*.

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(ii) outline how antibodies against *S. typhimurium* are produced.

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(b) The lumen of the intestinal tract contains nutrients and supports the growth of a diverse community of healthy bacteria that makes up the microbiota.

Suggest how the microbiota found in the lumen and in the outer mucus layer could help to protect against infection by this pathogenic bacteria.

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A novel *Salmonella* vaccine was developed by using a strain of *Salmonella* that has been rendered harmless. It is believed that newborns can be protected from *Salmonella* infections when their mothers are vaccinated during their pregnancy.

(c) Explain how vaccination of pregnant women can confer protection against *Salmonella* infections in newborns.

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(d) Explain why children need to be revaccinated although their mothers had been vaccinated during pregnancy.

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[Total:12]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answer must be set out in parts **(a)** and **(b)**, as indicated in the question.

4 Membranes and nucleic acids are commonly found in all living organisms.

(a) Discuss the importance of membranes to biological processes in the cell. [12]

(b) Prokaryotes and eukaryotes maintain the integrity of their genetic material to ensure that changes are minimal.

Outline why genetic stability is necessary in living organisms, and explain how genetic stability is maintained at the molecular, cellular and population levels. [13]

[Total: 25]

5 The cell is the basic unit of life, within which numerous biological processes that are essential to life occur.

(a) Discuss the importance of forming bonds that are weak and temporary during biological processes in the cell. [12]

(b) Cells are able to divide and give rise to new cells.

Cyclins are an important family of proteins that control the rate of cell cycle progression of a dividing cell. They are so named because, in general, the concentrations of the various cyclins change in a cyclical manner during the cell cycle. The changing levels of active cyclins thus ensure that there is sufficient time for various checks to be performed, to ensure that key cell cycle events have occurred normally.

Outline the checks made at cell cycle checkpoints, and explain how a dividing cell is able to **quickly** regulate the levels of cyclins which in turn control these checkpoints. [13]

[Total: 25]

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